

Informative Speech Outline

Name: Garrett Brownlee

Topic: The escalating prevalence of neurodegenerative diseases and preventative solutions

Organizational Pattern: Topical

Label and highlight your APA parenthetical citations (Last Name, Year).

Introduction

- A. **Attention Getter:** (get your audience interested in your speech): By the year 2050, the number of people living with dementia worldwide is projected to triple, reaching a staggering 153 million individuals. Have you ever considered how your current digital habits might shape your brain's health 40 years from now?
- B. **Relevance:** (state why this topic and information are relevant and valuable to your audience): While these diseases are often associated with the elderly, the foundations for prevention and the resulting societal economic burden are issues that will directly impact our generation as future professionals and caregivers.
- C. **Credibility:** (state your interest and why you are qualified to talk about this topic): This issue has been a topic of discussion in many of the neuroscience classes that I have taken: Affective Neuroscience, Advanced Principles, Cognition, Learning & Behavior (to name a few). This recurring topic always piqued my interest because of how differently it affects afflicted individuals, but also because...
- D. **Thesis:** (state your goal/purpose): As neurodegenerative diseases become an increasing global health concern, it is vital to understand the biological nature of these conditions, the modern lifestyle and environmental factors driving their rising prevalence, and the proactive preventative measures we can take to safeguard our cognitive health.
- E. **Preview:** (state your plan for the body; *In this speech I am going to...*): In this speech, I will examine the nature of neurodegenerative diseases, explore the environmental and technological factors driving their rise, and identify specific lifestyle habits we can adopt now to protect our long-term cognitive health.

Transition Sentence: (signal to the audience you're moving from the introduction to the first main idea): To understand why these diseases are on the rise, we must first look under the microscope at what is actually happening within the human brain.

Body

- I. **Main Point #1:** Neurodegenerative diseases are characterized by the progressive loss of structure or neuronal function, which leads to a significant decline in cognitive performance and physical health.
 - A. Walking through the biological mechanisms of decline
 - 1. These diseases involve the progressive accumulation of protein aggregates that cause toxicity and cell death.
 - 2. Think of this like a factory assembly line creating broken parts; eventually, these "misfolded" proteins clog the machinery of the brain until it can no longer function.

B. How does this disorder/process affect your health, all of you who are listening? Why should you care about neurodegenerative diseases?

1. Specific regions of the nervous system lose neurons, which manifests as symptoms like memory loss or impaired motor coordination.
2. For us as students, this is like a "short circuit" in a computer's wiring that prevents it from processing even simple commands.

C. How does this tie into the real world? Are there things we can do in our day-to-day lives to prevent/slow the onset of these devastating diseases?

1. ****research****
2. ****my interpretation of the research****

Transition Sentence: (signal to the audience you have finished Main Point #1; introduce Main Point #2):
Now that we understand the internal biological "short circuit," we must examine the external modern factors that are accelerating this process.

II. Main Point #2: The prevalence of these diseases is escalating due to an aging global population and modern lifestyle factors, including the impact of high-frequency AI and digital tool dependency on cognitive load.

A. The global impact of an aging population, a consequence of increasing life expectancy

1. The primary driver of rising dementia cases is the demographic shift toward an aging population structure. The second largest contributor is the extensive research and development given to diagnostic technology.
2. As modern medicine helps us to live longer, our brains are essentially "outliving" their natural functional lifespan.

B. How your technology use can contribute to neurodegenerative diseases

1. Research shows that heavy reliance on AI tools like ChatGPT can lead to "cognitive offloading," which reduces neural connectivity in areas responsible for memory and executive function.
2. Every time we delegate a complex thinking task to a machine, we are essentially letting our "mental muscles" atrophy.

C. In what ways can you intentionally moderate your technology use to prevent/slow disease onset?

1. Excessive AI use has been linked to a 32% reduction in relevant cognitive load and weakened critical thinking scores.
2. To prevent "digital dementia," we must shift from using technology as a replacement for thought to using it as a tool for active collaboration.

**Outline should be written in complete sentences.*

***Remember to incorporate all sources into the outline. Try your best to cite articles in a natural manner that fits the flow of the outline.*

****Highlight sentences with citations.*

*****Number of subpoints and sub-subpoints may vary. Just remember that if you have an A, you must have at least a B. If you have a 1, you must have at least a 2.*

Transition Sentence: (signal to the audience you have finished Main Point #2; introduce Main Point #3): Understanding these modern risks allows us to pivot toward the proactive steps we can take to build a more resilient brain.

III. **Main Point #3:** Individuals can take proactive steps toward prevention by maintaining "cognitive reserve" through specific lifestyle changes and mindful technology habits.

A. What is “cognitive reserve” and how do we build it?

1. Cognitive reserve is the brain's ability to improvise and find alternate ways of getting a job done, which can be improved through life-long learning and brain-challenging games.
2. Building this reserve is like "cross-training" your brain to ensure it has backup pathways if one "circuit" fails.

B. Physical and dietary interventions

1. Regular physical activity, such as walking 10,000 steps daily, is the single most accessible way to fight off neurodegeneration.
2. Moving your body isn't just for your heart; it pumps the nutrients and oxygen to your brain that are required for neural survival.

C. Being mindful of your technology habits

1. Conscious use of AI for repetitive tasks can free the brain for complex work, but it must be balanced with "deep work" zones free from digital interruptions.
2. The goal is intentional symbiosis—using technology to amplify our intelligence without losing the ability to think for ourselves.

Transitional Phrase: (signal to the audience you have finished all main points and will now conclude the speech): We have explored the biology, the risks, and the solutions, which brings us to a final reflection on our cognitive future.

Conclusion

- I. **Summary:** (Restate goal and main points): Today, we examined how neurodegenerative diseases "short circuit" the brain through protein misfolding, how our modern digital environment creates a "cognitive debt," and how we can use lifestyle changes to build a lasting cognitive reserve.
- II. **Memorable Close:** (Provide closure and leave a lasting impression): While we cannot yet fully stop the clock on aging, the power to protect the "human" in our brains rests in the choices we make today—starting with how we choose to engage with the world around us.

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References

- Alzheimer Society of Canada. (2024). *Brain-healthy tips to reduce your risk of dementia.*
- Alzheimer's Disease International. (2022). *New study predicts the number of people living with Alzheimer's disease to triple by 2050.*
- Frontiers in Aging Neuroscience. (2023). *Physical activity and lifestyle modifications in the treatment of neurodegenerative diseases.*
- MIT Media Lab / Psychology Today. (2025). *How ChatGPT May Be Impacting Your Brain.*
- The Lancet Public Health. (2022). *Estimation of the global prevalence of dementia in 2019 and forecasted prevalence in 2050.*

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